AINA MARYAM

Waxahachie, Texas | ainamaryam@utexas.edu | 469-732-8001 | github.com/aina-maryam

SKILLS

Tools and Libraries - Power BI, NumPy, Pandas, Plotly, Matplotlib, Seaborn, Scikit Learn , Microsoft Excel, Microsoft PowerPoint
Programming Languages - Python, C/C++, MATLAB (Familiar), R (Familiar), SQL (Familiar)
Data Analysis - Descriptive Statistics, Inferential Statistics, OMICS Data Analysis

EDUCATION

Master of Science in Artificial Intelligence — **August 2024 - Present**
University of Texas at Austin, TX

Master of Science in Bioinformatics | GPA 3.7 — **August 2013 - June 2015**
Maulana Azad National Institute of Technology, Bhopal, India

Bachelor of Science in Biotechnology | GPA 3.5 — **August 2007 - June 2011**
Shobhit University, Meerut, India

EXPERIENCE

Data Science Intern | Oeson, Remote, US — **August 2024 - Present**

- Conducted comprehensive exploratory data analysis on diverse datasets (public and proprietary) using Python libraries Pandas and NumPy.
- Utilized statistical methods to analyze complex data, informing business strategies and optimizing processes.
- Created informative, interactive visualizations using Tableau, Matplotlib, Plotly and Seaborn.
- Developed expertise in data storytelling, presenting findings to both technical and non-technical audiences.
- Implemented machine learning algorithms (supervised/unsupervised) using Scikit-learn.

Researcher | MANIT, Bhopal — **August 2014 - June 2015**

- Predicted disordered regions through hydrophobicity scores in proteins contributing to polyglutamine repeat disease development by neural network regression analysis.
- Created and curated extensive protein hydrophobicity datasets, encompassing ordered, disordered, and disease-associated proteins, leveraging UniProt, Disprot, PDB select__25 protein databases and ExpasyProtScale tool for data extraction and analysis.
- Trained a two-layer neural network using MATLAB's Neural Network Toolbox on a comprehensive dataset of ordered, disordered, and disease-related proteins, and created regression plots to visualize the relationship between predicted and actual outcomes.
- Translated research results successfully into accessible data visualizations using regression plots, facilitating the drawing of well-supported conclusions.

Student Trainee | IGIB - CSIR, New Delhi — **January 2011 - June 2011**

- Reviewed the molecular mechanisms of the SCA23 gene implicated in Spinocerebellar Ataxia type 23.
- Conducted DNA PCR and sequencing PCR of 64 samples using Automated Sanger Sequencer.
- Analyzed nucleotide sequences for novel point mutations and single nucleotide polymorphisms using the tool Lasergene DNASTAR Seqman.
- Presented research outcomes through clear and concise data visualizations, supporting evidence-based conclusions and decision making.

CERTIFICATIONS

Power BI for Beginners - SimpliLearn SkillUP | 2024

Tata Data Visualisation: Empowering Business with Effective Insights Job Simulation - Forage | 2024

- Completed a simulation involving transforming data and creating data visualizations using Power BI for Tata Consultancy Services.
- Created visuals for data analysis to help executives with effective decision making.

Revisiting Python Essentials - CodeSignal | 2024

Researcher's Guide to Omics Data Fundamentals - Fred Hutch Cancer Center - Coursera | 2024

Whole Genome Sequencing of Bacterial Genome - University of Denmark - Coursera | 2023

PUBLICATIONS

Published Articles on Medium - https://medium.com/@ainamaryam
Published in International Journal of Scientific Study 1 (3), 121-132
Classification & Molecular Biology of Orofaciodigital Syndrome Type I, Aina Maryam, Hansa Jain, Sanjyot Mulay | 2013

WORKSHOPS AND CONFERENCES

3rd IFIP International Conference on Bioinformatics IFIP-TC5 and Computer Society of India at MANIT, Bhopal | 2013

- Presented a paper on ‘Molecular genetics analysis of SCA type-23 in uncharacterized SCA patients of Indian Origin.’

Training Program on Bioinformatics - Department of Mathematics and Computer Applications at MANIT, Bhopal | 2014

- Conducted a lab on ‘Study of nucleotide and protein sequence databases, formats and retrieval’.
- Mentored students in biological data visualization and analytics, empowering them to effectively analyze and interpret complex biological data.